

Outbreak: Riverton

Scientific Response Director • Riverton

Every question in the campaign, with its answer.

15 missions 45 questions 13 callbacks 13 choice 12 sequence 7 diagnosis 7 protocol

6 ballpark written for **grade 12**

The scene is what the player reads before answering. *Why* and the rebuttals appear only afterwards, in the verdict — so the answer is never in the scene.

MISSION 1

The First Cluster

Open an investigation using a defensible provisional case definition and an explicit comparison baseline.

A false alarm could disrupt the city, but a delayed investigation could cost the only window for early containment.

1.1 Signal or noise? DIAGNOSIS EPIDEMIOLOGY & EVOLUTION

Returns as a callback on day 3

Emergency Department Network

Seven patients across three hospitals, four days, and the same four symptoms in every one of them. Morales has the surveillance printout, the reporting routes and the catchment figures spread out on the desk. The press office has asked twice whether it should say anything. Nothing goes out until somebody calls this.

ASSUMES a rate is a count measured against a population and a length of time · surveillance only sees the patients who reached it

Is this a signal, and what on the desk decides it?

Count	Cases meeting the syndrome	7 in four days
Baseline	Same syndrome	about 2 a year
Reporting	How the cases were found	5 by one alert rule
Denominator	Catchment population	supplied by all three hospitals
Season	Other respiratory syndromes this week	flat
Geography	Shared address	none found

✓ A real departure from the baseline

Seven in four days against two a year is far outside what the system usually does.

An artifact of the shared alert rule

Every case was found the same way

Ordinary autumn respiratory illness

This syndrome is common in autumn and the count is within normal seasonal movement.

Impossible to say — there is no denominator

Seven is a numerator

ANSWER

A real departure from the baseline

WHY

A count is not a signal. What makes it one is a comparison against something written down beforehand, and the baseline here is about two a year — so seven in four days is a departure of a couple of hundredfold, not a busy week. The reading that matters most is the quiet one: two of the seven were found on manual chart review, outside the alert rule. Had every case arrived through that one rule, the cluster could have belonged to the rule. Two found another way is what makes it real.

WHY THE OTHERS DO NOT HOLD

- ✗ The alert rule is the right worry, and it is the one this panel rules out. Five cases came through it and two did not, so the pattern survives being looked for a second way.
- ✗ Autumn respiratory illness would lift the other syndromes with it. The panel shows those flat, so whatever this is, it is not the season.
- ✗ The denominator is on the desk — all three hospitals supplied their catchment. That objection is the correct instinct and it has already been answered.

A count becomes a signal only when something says what was expected instead.

1.2 Build the provisional case definition

SEQUENCE

EPIDEMIOLOGY & EVOLUTION

PERSON STOP

Returns as a callback on day 4

Epidemiology Operations Room

Three hospitals start counting tomorrow morning, and each has drafted its own idea of what counts as a case. Whatever is on those three desks by the first shift is what the city's curve will be made of. Ortiz wants one definition, built in the right order.

ASSUMES a curve is only meaningful if every hospital counted the same thing · a test can be wrong by missing cases or by counting the wrong ones

Order the build so each step can be checked before the next one depends on it.

CORRECT ORDER

- 1 Review the first patient records for common features.
- 2 Specify observable symptoms, timing, and location.
- 3 Test the definition on known cases and obvious non-cases.
- 4 Publish the provisional definition with a revision date and uncertainty notes.

WHY

A case definition is an instrument, and like any instrument it is wrong in a known direction. Too narrow and the spread is invisible; too wide and the count fills with the ordinary respiratory illness of a city in autumn. That is why it is tested twice before it is issued — against patients already known to be cases, which catches a definition that is too narrow, and against obvious non-cases, which catches one that is too wide. A definition nobody tested is a curve nobody can interpret.

WHY THE OTHERS DO NOT HOLD

- ✗ Records first, because the shared features have to come out of the patients rather than out of the drafting. Specifying symptoms before reading the notes is writing down what you expected to find.
- ✗ Symptoms, timing and geography next, because those three are what a clinician at four in the morning actually applies. A definition that cannot be applied at speed is not an operational definition.
- ✗ Testing before publication, not after. Known cases catch a definition that is too narrow; obvious non-cases catch one that is too wide. Skip this and the first curve is uninterpretable in a direction nobody can name.
- ✗ The revision date goes out with the definition. Without it, three hospitals keep counting to a definition everybody knows is provisional and nobody is authorised to change.

Operational definitions convert vague concern into reproducible observation.

1.3 Spend the first response reserve

CHOICE

EPIDEMIOLOGY & EVOLUTION

PERSON STOP

Returns as a callback on day 7

Clinical Data Office

One reserve of response money, and three things the room wants to spend it on. Ortiz needs one of them started this afternoon; the other two wait until tomorrow at the earliest. Nothing measured so far would change a decision.

ASSUMES two sources can agree and still be wrong together if they were collected the same way

What does the office start this afternoon?

- ✓ A standardised line list across all three hospitals.

Interviews with every hospital director for an expert impression.

Wastewater sampling in the affected and comparison neighbourhoods.

A citywide warning, before transmission is understood.

ANSWER

A standardised line list across all three hospitals.

WHY

Three hospitals are about to produce three curves, and curves only add up if the same definition made them. The line list is what turns three sets of impressions into one set of observations with dates and denominators — and every later stream, the wastewater included, gets read against it. Start anywhere else and the first thing tomorrow's data has to be compared with does not exist yet.

WHY THE OTHERS DO NOT HOLD

- ✗ Interviews are fast and return impressions. An impression has no denominator, so it cannot say whether seven in four days is a departure from anything.
- ✗ Wastewater is the right second stream, and it is valuable precisely because it does not depend on who chose to seek care. It cannot name a patient or define a case, which is what this week needs.
- ✗ A warning issued before transmission is understood spends credibility on a hypothesis, and credibility is what every later message depends on.

Comparable observations come before independent ones, because everything else is read against them.

MISSION 2

What Kind of Agent?

Classify the broad agent category and identify the next discriminating test.

The wrong classification sends every later diagnostic and treatment effort down the wrong path.

2.1 What kind of agent fits the whole panel?

DIAGNOSIS

CLINICAL PHYSIOLOGY

Returns as a callback on day 5

Hospital Pathology Suite

Pathology has four results from the same positive specimens and will not put a name to any of it. The four are on the light box in front of you. Ortiz needs a class of agent before the diagnostic bench can choose a method at all.

ASSUMES cells and viruses differ in whether they can copy themselves alone · a negative control is a result, not an absence of one

Which explanation fits every reading, including the quiet negative controls?

Microscopy	Microscopy	No intact cellular forms seen	Several fields examined; image quality is good
Host-cell dependence	Host-cell dependence	Signal rises only when living host cells are present	Acellular control shows no increase
Bacterial marker	Bacterial marker	Not detected	Control bacterial sample is detected normally
Fungal cell-wall stain	Fungal cell-wall stain	Negative	Positive stain control is normal
Extraction blank	Extraction blank	Negative	No target signal in the blank

✓ An RNA virus-like agent

Not a cell of its own, and it can only copy itself inside a living host cell.

A bacterium of some kind

A cellular organism, which would leave bacterial markers and structures behind.

A fungus of some kind

A cellular organism with a cell wall the stain is built to find.

Laboratory contamination

Target material loose in the workflow rather than anything in the patients.

ANSWER

An RNA virus-like agent

WHY

The negatives are doing the work here. No cellular forms under the microscope, no bacterial marker, no fungal cell wall — three ways of being a cell, all absent. What is present is dependence: signal rises only where living host cells are, which is what an agent that cannot copy itself alone looks like. And the clean extraction blank is what stops the whole panel being about the laboratory instead of the patient.

WHY THE OTHERS DO NOT HOLD

- ✗ A bacterium is a cell and would leave cellular evidence — a marker, a wall, something under the microscope. Three separate looks for that came back empty.
- ✗ A fungus would show cell-wall material on the stain built to find it. That stain is negative, and it is a stain that rarely misses.
- ✗ Contamination is the right suspicion whenever a signal appears from nowhere, and it is checked by the blank. This blank is clean, so the signal came in with the specimens.

Agent classification should combine cellular structure, host dependence, molecular evidence, and controls rather than rely on one striking image.

2.2 From specimen to broad classification

SEQUENCE

EPIDEMIOLOGY & EVOLUTION

PERSON STOP

Microscopy Core

The microscopy core has patient material, healthy tissue and a negative control on the same bench, and four things it could do with them. Ortiz wants the order written down before anybody touches a specimen, because two of these consume what they measure.

ASSUMES a measurement of a contaminated preparation measures the contamination

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

CORRECT ORDER

- 1 Compare patient material with negative and healthy controls.
- 2 Characterize size, cellular structures, and nucleic-acid signature.
- 3 Ask whether the candidate reproduces independently or requires host cells.
- 4 Integrate the independent observations into a provisional classification.

WHY

Each step is only interpretable if the one before it held. Comparing against controls first is what separates the agent from the specimen it arrived in — measure a structure in a contaminated preparation and you have measured the contamination, at high magnification. Size and structure then say whether this is cellular at all, and only once you know that does asking whether it copies itself alone mean anything.

WHY THE OTHERS DO NOT HOLD

- ✗ Controls first, because everything after them is a claim about the patient rather than about the tube.
- ✗ Structure before behaviour: whether something needs a host cell is a question about an organism, and you do not have an organism until you know it is one.
- ✗ The replication question comes third because it is the slowest and the most specific — it is worth running only on a candidate the first two steps survived.
- ✗ Integration last. A provisional classification is what the other three produce, not a step you can take on its own.

A good classification is an evidence synthesis, not a label generated by one instrument.

2.3 Choose the next discriminating evidence CHOICE MOLECULAR DIAGNOSTICS

Returns as a callback on day 6

Molecular Identification Lab

Two hypotheses about the agent are still open and there is money for one more piece of work before the diagnostic bench has to commit. Three of the four things on the table would produce a result by tonight.

ASSUMES two methods that can fail the same way do not confirm each other

One experiment separates the two remaining hypotheses. Which?

✓ **Obtain a molecular signature using a validated broad panel.**

Repeat the same microscope image at higher magnification.

Compare growth in permissive and non-permissive cell systems.

Ask a recognised authority which answer is most likely.

ANSWER

Obtain a molecular signature using a validated broad panel.

WHY

The evidence so far is morphological, and the two candidates differ in their genome rather than their appearance. A molecular panel fails in a different way from microscopy, so agreement between the two would mean something; more microscopy could only agree with itself.

WHY THE OTHERS DO NOT HOLD

- ✗ Higher magnification is more of the evidence already held. If the first image could be misread, so can a sharper one taken the same way.
- ✗ The replication comparison is a real and independent test and it is what you run next. It answers whether the agent needs host cells — which the panel has already shown — rather than which candidate it is.
- ✗ An authority's opinion carries the authority's failure modes and adds no measurement at all.

Independent failure modes matter more than prestige or repeated views of the same evidence.

MISSION 3

The Point of Entry

Explain why one cell type is vulnerable and select the experiment that separates receptor binding from downstream replication.

A mistaken entry model could lead to a therapy that blocks the wrong stage while vulnerable cells continue to fail.

3.1 Crossing the cellular boundary

SEQUENCE

CELL & MEMBRANE BIOLOGY

Returns as a callback on day 8

Cell Culture Wing

Only some cell types in the respiratory tissue are affected, and the culture wing has four cards describing what happens when something crosses into a cell. Nobody has put them in order yet, and the therapy meeting is at four.

ASSUMES a cell membrane is a selective barrier, not a wall

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

CORRECT ORDER

- 1 A surface molecule binds a compatible receptor.
- 2 The membrane bends or a fusion process begins.
- 3 The biological material enters the cell or an intracellular compartment.
- 4 Cellular machinery is redirected after entry.

WHY

The order is physical, and it is why the tissue pattern is predictable. Nothing enters until something on the surface is recognised, and a cell without the right protein cannot be infected however much material reaches it. Recognition comes first, the membrane rearranges second because that is what binding triggers, material crosses third, and everything the cell then does wrongly is downstream of a barrier that has already been crossed.

WHY THE OTHERS DO NOT HOLD

- ✗ Binding first: it is the step that decides which cells are even candidates, which is what makes some cell types vulnerable and others not.
- ✗ The membrane bends because something bound to it. Fusion is a consequence of recognition, not a step that happens on its own.
- ✗ Material crosses only after the membrane has rearranged — that is what the rearrangement is for.
- ✗ The hijacked machinery is last because it is an effect. A cell whose machinery is redirected has already been entered.

Membranes are selective interfaces whose proteins can determine cell vulnerability.

3.2 Choose the membrane mechanism

PROTOCOL

CELL & MEMBRANE BIOLOGY

PERSON STOP

Returns as a callback on day 9

Membrane Biology Lab

Four observations from the culture bench, each describing something moving across a membrane. Adeyemi wants each one named by its mechanism before the entry model goes any further, and she does not want the names guessed from the molecules.

ASSUMES a concentration gradient has a direction · ATP is the cell's energy currency

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

CORRECT MATCHING

A small nonpolar molecule moves down its concentration gradient. → Simple diffusion.

An ion moves down its gradient through a selective protein pore. → Facilitated diffusion through a channel.

A solute moves uphill while ATP is consumed. → Active transport.

A large bound particle is enclosed by the membrane. → Endocytosis.

WHY

Three constraints tell these apart and none of them is the name of the molecule. Which way it goes: downhill needs no energy, uphill does, and the ATP consumption is the receipt. Whether a protein is involved: a small nonpolar molecule crosses the lipid directly, an ion cannot and needs a pore. And how big it is: a bound particle too large for any channel is taken in by the membrane wrapping it.

WHY THE OTHERS DO NOT HOLD

- ✗ Downhill and needing no protein is simple diffusion — the molecule is small and nonpolar enough to cross the lipid itself.
- ✗ Downhill but through a selective pore is facilitated diffusion. Still no energy spent; the protein supplies the selectivity, not the push.
- ✗ Uphill with ATP consumed is active transport, and the energy cost is the thing that identifies it.
- ✗ A large bound particle enclosed by the membrane is endocytosis — a different process from anything crossing the bilayer.

Transport mechanisms are inferred from constraints, not memorized from molecule names alone.

3.3 Separate binding from entry

CHOICE

CELL & MEMBRANE BIOLOGY

PERSON STOP

Returns as a callback on day 10

Structural Biology Room

Some cell types are infected and some are not, and nobody can say at which step the resistant ones stop the process. Structural biology has bench time for one experiment before the therapy target is chosen.

ASSUMES a chain of steps fails at its earliest broken link

Which experiment locates the step where resistant cells stop the process?

✓ Measure receptor abundance on vulnerable and resistant cells.

Track whether bound material is internalised.

Measure a late cellular response in both cell types.

Increase the exposure until every cell shows damage.

ANSWER

Measure receptor abundance on vulnerable and resistant cells.

WHY

Entry is a chain and the earliest divergence is the cause; everything after it is a consequence. Receptor abundance tests the first step, and it is the only measurement whose result changes what the next experiment should be.

WHY THE OTHERS DO NOT HOLD

- ✗ Internalisation is the right second experiment and it is only interpretable once you know both cell types can bind. Run it after, not instead.
- ✗ A late response tells you something already went wrong upstream, which is what you knew before you started.
- ✗ Flooding the system until everything is damaged destroys the difference between the two cell types, which is the entire measurement.

Mechanistic experiments should identify the earliest point at which two conditions diverge.

MISSION 4

Hijacked Cells

Build the information-flow model and choose a measurement that distinguishes more RNA from more efficient translation.

A rushed interpretation could make the team target a gene whose RNA is merely a consequence rather than the cause of cell damage.

4.1 From genetic information to protein

SEQUENCE

MOLECULAR DIAGNOSTICS

Returns as a callback on day 11

Sequencing Center

Affected cells are making RNA and proteins the sequencing centre does not recognise. Four cards describe how a cell gets from a gene to something that does a job. Nobody can say where the change is until the chain is on the board in order.

ASSUMES genes are read into RNA and RNA is read into protein

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

CORRECT ORDER

- 1 A DNA template is made accessible.
- 2 RNA polymerase produces an RNA transcript.
- 3 A ribosome reads codons in the messenger RNA.
- 4 The amino-acid chain folds and performs a cellular function.

WHY

Abundance can change at any one of these stages independently, which is exactly why the chain is worth stating rather than treating expression as one event. The template has to be reachable before anything transcribes it; a transcript has to exist before a ribosome can read it; and a chain of amino acids does nothing at all until it folds. Each stage is a separate place for the answer to be hiding.

WHY THE OTHERS DO NOT HOLD

- × Access first: a template nothing can reach is transcribed by nothing, whatever else is true of the cell.
- × Transcription second, because the message is what the next stage reads.
- × Translation third — a ribosome needs a transcript, and codons are read only once there is something to read them from.
- × Folding last, and it is not a formality: an unfolded chain is present, measurable, and useless.

Changes in protein abundance can arise at several distinct stages of gene expression.

4.2 Locate the molecular change

PROTOCOL

MOLECULAR DIAGNOSTICS

PERSON STOP

Returns as a callback on day 12

Gene Expression Lab

Four patterns from the gene-expression bench, each pairing a measurement with another measurement. Adeyemi wants each one placed at the stage it reports on, before anybody proposes a gene worth targeting.

ASSUMES the amount of a protein and the amount of its message are measured separately

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

CORRECT MATCHING

RNA rises while protein stays constant.	→ Possible translation limit or rapid protein turnover.
RNA is unchanged while protein rises.	→ Possible increased translation efficiency or reduced protein degradation.
A codon changes and one amino acid is substituted.	→ A missense sequence change.
Protein amount is normal but activity is low.	→ Possible folding, modification, or active-site defect.

WHY

RNA abundance, protein abundance, sequence and activity are four different measurements, and the central dogma is a chain of transformations rather than a promise that they move together. Message up with protein flat means the message is not being turned into protein, or the protein is being destroyed as fast as it is made. Protein up with message unchanged means the opposite end. And a protein present in normal amounts but not working is a claim about the molecule, not about how much of it there is.

WHY THE OTHERS DO NOT HOLD

- ✗ RNA rising while protein stays flat points downstream of transcription — a translation limit, or rapid turnover of the protein once made.
- ✗ Protein rising on unchanged RNA points to more efficient translation or slower degradation. Nothing about the gene has changed.
- ✗ One codon changed and one amino acid substituted is a sequence change, and it is visible only in the sequence.
- ✗ Normal amount with low activity is about folding, modification or the active site — the molecule is there and cannot do the job.

The central dogma is a chain of measurable transformations, not a claim that every stage changes together.

4.3 Find the controlling stage

CHOICE

MOLECULAR DIAGNOSTICS

PERSON STOP

Returns as a callback on day 13

Protein Analysis Core

An unfamiliar protein is abundant in affected cells. The protein analysis core has money for a few measurements, and the team is already arguing about which gene to target.

ASSUMES a protein's abundance is set by how fast it is made and how fast it is destroyed

Abundance alone fits every mechanism. Which measurement do you make first?

✓ Quantify the protein's messenger RNA across time.

Measure ribosome association with that RNA.

Measure the protein's half-life and degradation rate.

Sequence the neighbouring chromosomal region.

ANSWER

Quantify the protein's messenger RNA across time.

WHY

An abundant protein is consistent with more transcript, more efficient translation, or slower destruction. Transcript abundance is the one measurement that splits the three: if the message did not move, transcription is out and the cause is downstream of it.

WHY THE OTHERS DO NOT HOLD

- ✗ Ribosome association asks whether translation changed, and it only means something once you know whether the amount of message changed underneath it.
- ✗ Half-life asks whether the protein is simply being destroyed more slowly. A real possibility, and the third measurement rather than the first.
- ✗ The neighbouring region answers a question nobody has asked about a protein whose gene is already identified.

Mechanism requires measurements that span the full causal chain.

MISSION 5

A Test Before Morning

Approve a diagnostic workflow only after its analytical and clinical limitations are explicit.

An unreliable test could either miss contagious patients or isolate thousands of healthy people.

5.1 Build a trustworthy amplification run

SEQUENCE

MOLECULAR DIAGNOSTICS

Returns as a callback on day 14

Sample Processing Room

Hospitals want a diagnostic by sunrise and the sample-processing room has four hours. Four steps are on the whiteboard. Amplification multiplies whatever is in the tube by a factor of a billion, which is the reason this order is argued about at all.

ASSUMES amplification copies whatever template is present, wanted or not

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

CORRECT ORDER

- 1 Extract material while preserving sample identity.
- 2 Include positive, negative, and no-template controls.
- 3 Amplify the target with validated primers and cycling conditions.
- 4 Interpret the patient signal only if all controls behave as expected.

WHY

A patient result is the output of a controlled process, and read before the controls are read it is an instrument reading rather than evidence. Identity has to survive extraction or the result belongs to no one. The controls have to be in the run, not added afterwards — a positive control shows the chemistry worked, and a no-template control shows the signal did not come from the room. Only when they behave is the patient well worth interpreting.

WHY THE OTHERS DO NOT HOLD

- ✗ Extraction first, with identity preserved. A result that cannot be tied to a patient is not a diagnosis of anybody.
- ✗ Controls go into the run alongside the samples. Added later they test a different run from the one that produced the result.
- ✗ Amplification third, with validated primers — the step everything else exists to make interpretable.
- ✗ Interpretation last and conditional. If a control misbehaved, the patient wells are uninterpretable no matter how clean they look.

A diagnostic result is the output of a controlled process, not simply a machine reading.

Returns as a callback on day 15

PCR Diagnostics Lab

The overnight run flagged several patients positive. The controls and a second platform ran alongside them, and all of it is on the screen in front of you. Hospitals are waiting to isolate on this result.

ASSUMES a no-template control contains water and no sample · two methods with different chemistry fail in different ways

Which explanation best fits the patient results and all of the controls?

Patient wells	Patient wells	Target signal detected	Several patient samples cross the detection threshold
No-template control	No-template control	Target signal detected	This control should contain no target material
Positive control	Positive control	Detected at expected level	Core amplification chemistry is functioning
Independent platform	Independent platform	Patient target not detected	Uses a different detection method
Instrument baseline	Instrument baseline	Stable	No broad detector drift is present

True patient positives

The patient samples contain the target and the controls are behaving normally.

✓ Carryover contamination

Target material entered part of the amplification workflow, making samples and the no-template control positive.

Reagent failure

The reaction chemistry failed, so true target should not amplify reliably.

Sample inhibition

Something in patient material suppresses amplification and tends to create false negatives.

ANSWER

Carryover contamination

WHY

A control is part of the evidence, not a decoration on it. The no-template control is water — there is nothing in it to amplify — so a target signal there means target material is loose in the workflow, and anything the workflow touched is suspect. The positive control working rules out failed chemistry, and an independent platform not finding the target in the same patients rules out the patients. One quiet well overturns an impressive-looking plate.

WHY THE OTHERS DO NOT HOLD

- ✗ True positives would not put a signal in a tube containing no sample. That well is the one reading that cannot be explained by infected patients.
- ✗ Reagent failure would stop true target amplifying, and the positive control amplified exactly as expected.
- ✗ Inhibition suppresses amplification and produces false negatives. This plate has the opposite problem.

A control is part of the evidence, not a decoration: a positive blank can overturn an apparently impressive patient signal.

5.3 What does a positive mean?

BALLPARK

EPIDEMIOLOGY & EVOLUTION

Clinical Statistics Desk

The mayor's office wants to screen a low-prevalence population and treat every positive as a case. The clinical statistics desk has the assay's performance figures and the size of the group in front of it.

ASSUMES prevalence is the share of a population that has the condition · sensitivity and specificity are properties of a test, not of a population

Estimate the chance that a positive result is a true case.

GOVERNING RELATIONSHIP Of everyone tested, the true positives are prevalence × sensitivity; the false positives are (1 – prevalence) × (1 – specificity). The answer is true positives as a share of all positives.

10,000 people screened

0.01 (prevalence in the screening group)

0.90 (sensitivity)

0.99 (specificity)

0.30 (prevalence on the affected ward)

0.10 (the fraction reporting symptoms)

$$10,000 \text{ people screened} \times 0.01 \times 0.90 \div (10,000 \text{ people screened} \times 0.01 \times 0.90 + 10,000 \text{ people screened} \times (1 - 0.01) \times (1 - 0.99)) \times 100$$

ANSWER

90 true positives against 99 false ones, so about 48% of positives are real.

WHY

Specificity is a property of the test; the fraction of positives that are real is a property of the test and the population together. Moving this assay from the ward to the city changes nothing about the chemistry and almost everything about what a positive means.

Clinical meaning depends on both test performance and the population being tested.

MISSION 6

Why Are Some Patients Sicker?

Identify the physiological bottleneck and select measurements that distinguish ventilation, diffusion, circulation, and cellular oxygen use.

The wrong physiological diagnosis could intensify a treatment that cannot reach the true failing organ system.

6.1 Locate the oxygen-delivery failure

DIAGNOSIS

CLINICAL PHYSIOLOGY

Intensive Care Unit

Two patients arrived with the same complaint and are going in opposite directions. The intensive care unit has a full cardiopulmonary panel on one of them, and the team is about to escalate a treatment on the strength of it.

ASSUMES oxygen travels air to lung to blood to tissue, and each handover can fail separately

Which bottleneck fits the entire cardiopulmonary panel?

Arterial oxygen saturation	Arterial oxygen saturation	86%	Lower than expected
End-tidal carbon dioxide	End-tidal carbon dioxide	39 mmHg	Near the expected resting range
Cardiac output	Cardiac output	5.2 L/min	Near the expected adult resting range
Hemoglobin	Hemoglobin	14 g/dL	Near the expected adult range
Lung imaging	Lung imaging	Diffuse gas-exchange abnormality	No evidence of a primary pump failure

Ventilation failure

Too little fresh air reaches the alveoli; carbon dioxide would often rise along with falling oxygen.

✓ Gas-exchange failure

Air reaches the lungs, but oxygen transfer from alveoli into blood is impaired.

Circulatory delivery failure

Oxygenated blood is present but cardiac output is too low to deliver it to tissues.

Anemia

Blood flow and lung exchange may be adequate, but too little hemoglobin limits oxygen carriage.

ANSWER

Gas-exchange failure

WHY

Oxygen delivery is a chain, and normal values at one link are not reassurance — they are evidence that the failure is elsewhere. Carbon dioxide near normal says air is reaching the alveoli, so ventilation is working. Cardiac output normal says the pump is moving blood. Haemoglobin normal says the blood can carry oxygen. What is left is the handover between air and blood, and the imaging says the same thing.

WHY THE OTHERS DO NOT HOLD

- ✗ Ventilation failure would usually push carbon dioxide up as oxygen falls, because the same failing airflow carries both. This carbon dioxide is near normal.
- ✗ Circulatory failure means oxygenated blood that never arrives, and cardiac output here is 5.2 L/min. The blood is moving.
- ✗ Anaemia limits how much oxygen each litre can carry, and haemoglobin is 14 g/dL. The carriage is not the problem.

Oxygen delivery is a chain. Normal measurements at one link can be powerful evidence that the failure lies somewhere else.

6.2 Estimate oxygen delivery

BALLPARK

EPIDEMIOLOGY & EVOLUTION

Cardiopulmonary Physiology Lab

The bedside argument has stalled on which of two patients is worse off. Both have the same oxygen content per litre of blood; one moves three litres a minute and the other five.

ASSUMES a rate is an amount multiplied by how fast it is delivered

Estimate oxygen delivery for the patient with the lower cardiac output.

GOVERNING RELATIONSHIP Oxygen delivered = blood flow × oxygen carried per unit of blood.

3 L/min (cardiac output, first patient)

0.20 L O₂/L (oxygen content of the blood)

5 L/min (cardiac output, comparison patient)

86 % (arterial saturation)

14 g/dL (haemoglobin)

$$3 \text{ L/min} \times 0.20 \text{ L O}_2/\text{L}$$

ANSWER

$3 \times 0.20 = 0.6 \text{ L O}_2/\text{min}$, against 1.0 for the comparison patient.

WHY

Two patients with identical saturations can differ by a factor of two in delivery, because the flow term is doing the work. A concentration is not a rate.

A normal concentration does not guarantee a normal delivery rate; flow matters.

6.3 Measure the failing link

CHOICE

CLINICAL PHYSIOLOGY

PERSON STOP

Clinical Chemistry Bench

A patient is deteriorating and the team wants to commit to a treatment. There is time and money for one more measurement at the bedside before it does.

ASSUMES a failure shows itself at the point in a chain where the numbers stop matching

One bedside measurement before the team commits to a treatment. Which?

✓ **Tissue lactate and acid-base status.**

Arterial oxygenation and carbon dioxide.

Cardiac output or a validated flow surrogate.

The same pulse oximeter reading, repeated on three devices.

ANSWER

Tissue lactate and acid-base status.

WHY

Oxygen delivery is a chain and the panel has already measured its top: saturation is low, carbon dioxide and cardiac output are not. What nobody has measured is the far end — whether delivery is actually failing to meet demand in the tissues, which is the thing the treatment is meant to change.

WHY THE OTHERS DO NOT HOLD

- ✗ Arterial oxygenation and carbon dioxide are already on the chart from the earlier panel. Measuring them again is precision about a number that is not in dispute.
- ✗ Cardiac output reports the pump, and the pump has already read normal. Worth repeating if the patient changes; not the missing measurement now.
- ✗ Three identical devices agreeing tells you the devices agree. It does not add an independent observation.

Integrated physiology requires evidence from multiple points along the transport chain.

MISSION 7

The Immune System Turns

Construct a timeline of immune responses and decide whether deterioration reflects persistent agent burden, delayed adaptive response, or excessive inflammation.

Suppressing a protective response or ignoring immune-mediated injury can both be fatal.

7.1 Build the immune response timeline

SEQUENCE

IMMUNOLOGY & TREATMENT

Immunology Lab

Patients are improving and then getting worse again, and the immunology lab wants the normal response on the board before it says which part has gone wrong. Four cards, no order yet.

ASSUMES the immune system has a fast general response and a slower specific one

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

CORRECT ORDER

- 1 Innate sensors recognize damage or unfamiliar molecular patterns.
- 2 Local signals recruit and activate immune cells.
- 3 Antigen-specific lymphocytes expand and act.
- 4 A smaller memory population persists after resolution.

WHY

The phases are ordered by speed and by specificity, and that ordering is what makes a clinical measurement interpretable. Innate sensors fire within hours because they recognise patterns rather than a particular agent. Local signals then recruit cells to where the damage is. Antigen-specific lymphocytes take days because they have to find the right clone and expand it. Memory is what persists after — a smaller population, kept.

WHY THE OTHERS DO NOT HOLD

- ✗ Innate recognition first: it needs no prior exposure, which is exactly why it can be the fastest.
- ✗ Recruitment second — signals bring cells to the place, and nothing specific can act until cells are there.
- ✗ Clonal expansion third, and it is slow because finding and multiplying the right lymphocyte takes days.
- ✗ Memory last. It is what is left after resolution, and it is the reason a second exposure goes differently.

Innate and adaptive immunity are coordinated phases, not competing systems.

7.2 Protective response or runaway inflammation?

DIAGNOSIS

CLINICAL PHYSIOLOGY

Intensive Care Unit

Several patients began to worsen after the amount of detectable pathogen had already started to fall. The panel from the last forty-eight hours is on the screen, and the ward wants to know what it is treating.

ASSUMES the immune response can damage tissue while it clears an infection

Which mechanism best explains the reversal in the full panel?

Pathogen signal	Pathogen signal	Falling over 48 hours	Lower than at admission
Inflammatory cytokine panel	Inflammatory cytokine panel	Rising sharply	Multiple inflammatory signals increase together
Neutralizing antibody	Neutralizing antibody	Present	Adaptive response is active
Oxygenation	Oxygenation	Worsening	Physiology deteriorates despite lower pathogen signal
Bacterial cultures	Bacterial cultures	No growth	No evidence for a new bacterial infection

Uncontrolled pathogen replication

The agent is multiplying faster and directly increasing tissue injury.

✓ **Dysregulated inflammatory injury**

The immune response remains strongly activated and damages tissue even as pathogen burden falls.

New bacterial superinfection

A second bacterial infection is now driving the decline.

Measurement noise

The apparent changes are unrelated fluctuations without a shared mechanism.

ANSWER

Dysregulated inflammatory injury

WHY

Timing is the discriminator. If the agent were driving this, its curve and the damage would rise together — instead the pathogen signal is falling while inflammatory signals climb and the patient gets worse. That is a response that has outlived its cause. Neutralising antibody being present says the specific response has arrived, and the negative bacterial cultures remove the rival that would otherwise explain a late deterioration.

WHY THE OTHERS DO NOT HOLD

- ✗ Uncontrolled replication would show a rising pathogen signal. This one is falling, over forty-eight hours, while the patient declines.
- ✗ A bacterial superinfection is the classic late deterioration, and it is the reason cultures were sent. They are negative.
- ✗ Measurement noise does not move four readings in a coherent direction at once, and the antibody result is not noise.

A biological response can become part of the problem. The strongest explanation must account for timing as well as the direction of each measurement.

7.3 Treat the mechanism, not the marker

CHOICE

CLINICAL PHYSIOLOGY

PERSON STOP

Pathology Archive

Late deterioration is killing patients and the ward wants to start immunosuppression tonight. The pathology archive has the samples to answer why, and answering takes hours the ward says it does not have.

ASSUMES suppressing a response removes it whether or not it is the thing causing harm

The ward wants to start immunosuppression tonight. What do you do?

✓ **Measure agent burden and inflammatory markers together over time first.**

Suppress the immune response immediately in everyone deteriorating.

Compare immune-cell states in improving and worsening patients.

Send pathology on affected tissue to identify what is being damaged.

ANSWER

Measure agent burden and inflammatory markers together over time first.

WHY

The two candidate drivers — a pathogen still replicating and a host response that has become the injury — are distinguished by which curve is rising while the other falls. Paired measurements over time separate them within hours, and the answer decides whether suppression helps or removes the thing that is clearing the infection.

WHY THE OTHERS DO NOT HOLD

- ✗ Suppressing now treats one hypothesis as settled. If agent burden is still rising in some of these patients, it removes their only defence.
- ✗ Comparing immune-cell states is the right study and returns in days. The ward is asking about tonight.
- ✗ Pathology says where the damage is. It does not say whether the pathogen is still driving it, which is the question the treatment turns on.

Intervention should follow evidence about what the immune system is doing, where, and when.

MISSION 8

The Hidden Transmission Route

Identify the hidden route and target surveillance where it changes decisions rather than where sampling is easiest.

If transmission begins before severe symptoms, a hospital-only response will always be late.

8.1 Match the surveillance signal

PROTOCOL

IMMUNOLOGY & TREATMENT

Wastewater Treatment Plant

Hospital isolation has held for nine days and cases keep appearing. Four data streams are on the table at the wastewater plant, and each one saw a different slice of the city.

ASSUMES who ends up in a dataset is decided by how the data was collected

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

CORRECT MATCHING

Wastewater concentration rises before hospital admissions.	→ Possible early population-level signal, but not individual diagnosis.
Household interviews show several exposures before symptoms.	→ Evidence that infectious contact may precede symptoms.
A transit survey samples only volunteers at one station.	→ Potentially biased sample that may not represent riders.
Hospital prevalence remains high after new admissions decline.	→ Long illness duration can keep prevalence high after incidence falls.

WHY

What a stream can reveal is fixed by who is observed, when, and how they came to be observed. Wastewater sees everyone connected to the sewer whether or not they sought care, which makes it early and anonymous at the same time. Household interviews reach only households that agreed. A volunteer survey at one station samples volunteers at one station. And prevalence counts everybody currently ill, so it stays high while people are still ill even after new cases stop.

WHY THE OTHERS DO NOT HOLD

- ✗ Wastewater rising before admissions is a population signal that arrives early and can never name a patient.
- ✗ Exposures found before symptom onset are evidence about when transmission happens, from the only stream that asks people directly.
- ✗ A volunteer sample at one station is a sample of volunteers at one station — the bias is in who agreed, not in the measurement.
- ✗ Prevalence stays high after incidence falls when the illness lasts a long time. The two are different quantities and this is how they come apart.

Transmission inference depends on who is observed, when, and through which sampling process.

8.2 Can the cluster grow?

BALLPARK

ECOLOGY & ONE HEALTH

PERSON STOP

Field Epidemiology Office

About twenty people are infectious now, and the field office's best estimate is that each is producing roughly 1.4 more. Nobody has written down what that means for next week.

ASSUMES a chain grows when each case produces more than one new case

Estimate how many infections the next generation produces.

GOVERNING RELATIONSHIP Next generation = current cases × the average number each one infects.

20 people currently infectious

1.4 new infections per person per generation

5 days (the serial interval)

140 cases reported so far

0.7 new infections per person, after control

20 people currently infectious × 1.4 new infections per person per generation

ANSWER

$20 \times 1.4 \approx 28$, so the chain grows.

WHY

Twenty-eight against twenty is growth, and the fact that it is growth — not the size of the number — is what decides whether the chain burns out or doubles every fortnight.

A threshold near one can determine whether small changes produce growth or decline.

8.3 Find transmission before the hospital

CHOICE

EPIDEMIOLOGY & EVOLUTION

Transit and Mobility Center

Nine days of hospital isolation, and cases keep appearing in neighbourhoods with nobody admitted from them. There is enough money this week for one of the four things the room has proposed.

ASSUMES a measurement taken downstream of a process reports it after it has happened

Cases appear in neighbourhoods with nobody admitted. What do you commission?

✓ Expand wastewater sampling across the connected sewersheds.

Conduct structured household and workplace contact studies.

Add more hospital bed and admission counts.

Sample one comparison neighbourhood with no reported cases.

ANSWER

Expand wastewater sampling across the connected sewersheds.

WHY

Hospital isolation has held for nine days, so the chain is running somewhere the hospital cannot see. Wastewater reports the population rather than the patients who reached care, and sampling it by sewershed turns 'somewhere' into a map — which is the thing every other option needs and none of them supplies.

WHY THE OTHERS DO NOT HOLD

- ✗ Contact studies are how presymptomatic transmission is established, and they need a place to start. Which households, in which neighbourhoods, is what the sewershed map is for.
- ✗ More bed counts measure the outcome you are trying to get ahead of. They will confirm the problem after it arrives.
- ✗ A comparison neighbourhood is worth sampling and is part of the same programme, not a substitute for finding where transmission is happening.

The most useful surveillance reaches upstream of the outcome it is trying to prevent.

MISSION 9

The Animal Connection

Build a defensible reservoir hypothesis using ecological exposure, prevalence, and genetic relatedness.

A mistaken reservoir declaration could trigger ecological damage while the true source continues seeding cases.

9.1 Reservoir, vector, or incidental host?

PROTOCOL

ECOLOGY & ONE HEALTH

One Health Field Station

Sequences from three patients sit close to a virus found in animals along the river corridor. The field station has four ecological patterns on the board and a trapper waiting outside with more animals.

ASSUMES a species can carry an agent, spread it, or merely be near it

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

CORRECT MATCHING

A species carries the agent persistently with little disease. → Potential reservoir host.

A biting organism transfers the agent between hosts. → Potential biological or mechanical vector.

A species becomes ill but rarely transmits onward. → Incidental or dead-end host.

A species is common near cases but consistently tests negative. → Ecological association without evidence of infection.

WHY

Ecological role is a claim about dynamics, not about proximity or about which animal people are frightened of. A reservoir carries the agent persistently and stays well enough to keep carrying it. A vector moves it between hosts without being the place it persists. A species that sickens and dies without passing it on is where transmission stops. And a species that is simply common near cases and never tests positive shares a habitat with people, which is not the same as sharing an infection.

WHY THE OTHERS DO NOT HOLD

- ✗ Persistent carriage with little disease is what a reservoir looks like — it has to survive the infection to maintain it.
- ✗ A biting organism that moves the agent between hosts is a vector whether or not it is ever the place the agent persists.
- ✗ A species that becomes ill and rarely transmits is a dead end. The chain reaches it and stops.
- ✗ Common near cases and consistently negative is a shared habitat, not a shared infection. Proximity is the weakest evidence on the board.

Ecological roles are inferred from dynamics, not appearance or proximity alone.

9.2 Test the reservoir hypothesis

SEQUENCE

ECOLOGY & ONE HEALTH

PERSON STOP

Wildlife Ecology Site

Trappers report sick animals on the flood plain, and nobody has yet sampled in a way that could tell a reservoir from a bystander that caught it from people. Four steps, and the campaign gets funded once.

ASSUMES where you look decides what you can find

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

CORRECT ORDER

- 1 Map human cases against habitat and exposure patterns.
- 2 Sample candidate species using an unbiased design and comparison sites.
- 3 Compare prevalence and genetic signatures across hosts.
- 4 Test whether the combined evidence explains repeated spillover.

WHY

A positive animal is not a reservoir. What turns it into one is a design that could have found the agent somewhere else and did not, which is why the order matters: exposure patterns say where to sample, and sampling built around comparison sites is what makes a prevalence figure mean anything. Only then do prevalence and genetic relatedness across hosts say whether this species is in the transmission cycle, and only then can repeated spillover be tested rather than assumed.

WHY THE OTHERS DO NOT HOLD

- ✗ Human cases against habitat first — that is what tells you which species and which places are even candidates.
- ✗ Unbiased sampling with comparison sites second. Find the agent only where you looked hardest and you have measured your own search.
- ✗ Prevalence and genetic relatedness third: they connect the two populations, and they need samples from both to compare.
- ✗ Repeated spillover last. One event can be a coincidence; a reservoir claim is a claim about a process that keeps happening.

A reservoir claim requires population and evolutionary evidence together.

9.3 Choose the One Health campaign

CHOICE

EPIDEMIOLOGY & EVOLUTION

Comparative Genomics Lab

There is one field campaign to fund and its design decides what can be concluded from it. Trappers, a genomics bench and a nervous agriculture department are all waiting on the answer.

ASSUMES a sample tells you about the population it was drawn from and no other

One field campaign. What is its design?

✓ **Stratify sampling across species, habitat, season and comparison sites.**

Sample the most feared species intensively near one case household.

Sequence every positive sample from humans and animals.

Remove wildlife from the flood plain while the investigation continues.

ANSWER

Stratify sampling across species, habitat, season and comparison sites.

WHY

A reservoir claim is a claim about prevalence, and prevalence only means something against a design that could have found the agent somewhere else. Comparison sites are what separate a reservoir from a bystander that caught it from people.

WHY THE OTHERS DO NOT HOLD

- ✗ Sampling hard around one household guarantees a positive that cannot be generalised, and it will be the species people were already afraid of.
- ✗ Sequencing is essential and comes second: you can only sequence what the sampling design found, and a biased design produces a confident tree of the wrong animals.
- ✗ Removing wildlife before involvement is established destroys the evidence, and does nothing at all if the reservoir is a species nobody removed.

One Health decisions should connect human, animal, and environmental evidence before intervention.

MISSION 10

The Mutation

Distinguish change in frequency from proof of advantage and identify evidence that tests competing evolutionary explanations.

Misreading drift as selection can produce unnecessary alarm; missing a genuine advantage can leave the response behind the evolving population.

10.1 From variation to evolutionary change

SEQUENCE

MOLECULAR DIAGNOSTICS

Sequencing Center

A genetic variant is rising in one part of the city and the word evolution is being used in the briefing room. Four cards describe how a population changes. Nobody has put them in order.

ASSUMES offspring inherit traits from their parents

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

CORRECT ORDER

- 1 Heritable variants arise through mutation or existing diversity.
- 2 Environmental conditions cause variants to differ in reproductive success.
- 3 Successful variants contribute disproportionately to later generations.
- 4 Population frequencies change over generations.

WHY

Nothing in this chain says a rising variant must be advantageous, and that is the point of laying it out. Variation has to exist before anything can act on it — selection does not create the variant it needs when it needs one. Conditions then make some variants reproduce more than others, those variants leave more descendants, and the frequency moves. Take away the middle step and frequencies still move, for reasons that have nothing to do with fitness.

WHY THE OTHERS DO NOT HOLD

- ✗ Variation first, and it arises independently of whether it is useful. This is the step people reverse.
- ✗ Differential reproductive success second: the environment does the selecting, on variation that was already there.
- ✗ Successful variants leaving more descendants is the mechanism, and it happens over generations rather than within one.
- ✗ Changed frequencies last. They are the observation, and every other card is the explanation for it.

Natural selection changes frequencies through heritable differences in reproductive success.

10.2 Selection or sampling artifact?

DIAGNOSIS

EPIDEMIOLOGY & EVOLUTION

Evolutionary Dynamics Group

A variant has gone from 8 to 31 per cent of sequences in four weeks and the briefing is in an hour. The evolutionary dynamics group has run its checks and put them all on one screen.

ASSUMES a change in who gets tested changes what the tests find

Which explanation best fits the frequency trend and the independent checks?

Variant frequency	Variant frequency	8% -> 31% in four weeks	Rise appears in the combined dataset
Independent hospitals	Independent hospitals	Same upward trend in three systems	Different collection teams reproduce the pattern
Sequencing depth	Sequencing depth	Stable	No systematic loss of low-count samples
Neutral control variants	Neutral control variants	No coordinated rise	Background frequencies remain stable
Adjusted growth estimate	Adjusted growth estimate	Variant lineages grow faster after place and date are controlled	Comparison reduces a major sampling confounder

✓ Real selective advantage

The variant has a reproducible growth advantage and increases in frequency across independent populations.

Changing sampling mix

The apparent rise is created because later samples come from different places or patient groups.

Sequencing artifact

A technical change makes the variant easier to detect over time.

Single founder event only

One early cluster happens to contain the variant, with no broader growth advantage.

ANSWER

Real selective advantage

WHY

A frequency change is an observation; selection is one explanation for it, and it competes with founder effects, with a change in who is being sampled, and with a technical shift. Each of those is ruled out by a specific check here. Three independent hospital systems showing the same rise is not one cluster. Stable sequencing depth is not a technical change. Neutral variants staying flat means the whole population was not resampled. And the rise survives adjustment for place and date.

WHY THE OTHERS DO NOT HOLD

- ✗ A changing sampling mix would move the neutral control variants too, because they ride on the same samples. Those are flat.
- ✗ A sequencing artefact would show as changing depth or coverage, and the depth is stable across the four weeks.
- ✗ A single founder cluster would not repeat across three independent hospital systems, and it would not survive adjusting for place.

A frequency change is an observation; selection is an explanation that requires controls against sampling, founder effects, and technical drift.

10.3 Track a rising variant

BALLPARK

EPIDEMIOLOGY & EVOLUTION

PERSON STOP

Regional Surveillance Hub

Before anybody argues about why the variant is rising, the surveillance hub needs the rise itself. Week one and week three sequenced different numbers of samples.

ASSUMES a fraction needs both a numerator and the denominator it came from

Estimate the change in variant frequency, in percentage points.

GOVERNING RELATIONSHIP A share = the count ÷ the number sequenced. Take the share in each week, then the difference.

40 variant, week 1

400 sequenced, week 1

180 variant, week 3

600 sequenced, week 3

220 variant, weeks 1 and 3 together

1,000 sequenced across both weeks

$(180 \text{ variant, week 3} \div 600 \text{ sequenced, week 3} - 40 \text{ variant, week 1} \div 400 \text{ sequenced, week 1}) \times 100$

ANSWER

10% in week one, 30% in week three: an increase of 20 percentage points.

WHY

A frequency is a ratio, and the ratio only means something if the two denominators were produced the same way. A change in who gets sequenced moves the numerator with nothing happening in the population.

Evolutionary inference begins with accurate frequencies and then tests causal explanations.

MISSION 11

Treatment Failure

Separate inherited resistance from non-genetic treatment failure and propose a strategy that reduces selective advantage without undertreating patients.

A blunt response could expose patients to toxicity while strengthening the very resistant population the therapy is meant to suppress.

11.1 Why did the therapy fail? DIAGNOSIS EPIDEMIOLOGY & EVOLUTION

Clinical Ward

A therapy that worked is failing. The ward has the drug levels, the agent burden over time, the sequence at the target site and a reference isolate, and it wants to change everybody's prescription this afternoon.

ASSUMES a drug only works if it reaches the target at a high enough concentration

Which explanation fits the complete treatment-response panel?

Drug concentration	Drug concentration	Within expected therapeutic range	Measured repeatedly during treatment
Agent burden	Agent burden	Falls, then rebounds	Initial response is followed by renewed growth
Target-site sequence	Target-site sequence	Resistance-associated variant becomes dominant	Frequency rises after treatment begins
Reference isolate	Reference isolate	Remains drug-sensitive	The assay can still detect drug activity
Administration record	Administration record	Doses documented	No major missed-dose pattern

Poor drug delivery

The drug never reaches an effective concentration in the patient.

✓ Evolved resistance

Treatment favors a resistant subpopulation that later dominates and restores growth.

Wrong infectious agent

The original identification was incorrect, so treatment never targeted the cause.

Assay contamination

The rebound is an artificial laboratory signal unrelated to the patient.

ANSWER

Evolved resistance

WHY

Treatment failure has several causes and they are told apart by which measurements are normal. The drug reached therapeutic concentration and the burden fell, so it was working — that rules out delivery and dose. It then rebounded while the patient was still taking it, and the resistance-associated variant at the target site became dominant over the same period. The reference isolate staying sensitive is what says the assay and the drug are both still fine. Evolution is visible here on the timescale of a ward round.

WHY THE OTHERS DO NOT HOLD

- ✗ Poor delivery means the drug never arrives, and the concentration is inside the therapeutic range with an initial response to prove it.
- ✗ A misidentified agent would not have responded at all, and this one fell before it came back.
- ✗ Assay contamination would not spare the reference isolate, which was run alongside and remains sensitive.

Evolution can be observed in real time when a treatment changes which variants are most successful.

11.2 How resistance spreads

SEQUENCE

EPIDEMIOLOGY & EVOLUTION

PERSON STOP

Microbiology and Evolution Lab

The therapy worked for six weeks and now fails in one patient in four. Failures cluster on the wards where it has been in use longest, and somebody in the meeting has said the organism learned to resist it.

ASSUMES a population contains variation before anything acts on it

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

CORRECT ORDER

- 1 A population contains rare variants with differing susceptibility.
- 2 Treatment removes susceptible individuals more effectively.
- 3 Less-susceptible variants survive and reproduce.
- 4 The treated population becomes enriched for resistance.

WHY

Treatment does not teach an organism anything, and it does not create the variant it needs. The variation is already there, rare, before the drug arrives. What the drug changes is which variants leave descendants: it removes the susceptible ones more effectively, the less susceptible survive and reproduce, and the treated population ends up enriched for resistance. The clustering on long-treated wards is that process having run longer, not the organism having learned faster.

WHY THE OTHERS DO NOT HOLD

- × Rare variants exist first. This is the step that separates an evolutionary explanation from a story about the drug teaching the organism.
- × Treatment removes the susceptible preferentially — that is what selection is, and it acts on what is already present.
- × The survivors reproduce, which is how a rare variant stops being rare.
- × Enrichment last: it is the outcome, and it is what the ward is now measuring.

Selection acts on existing or newly arising variation by differential survival and reproduction.

11.3 Recover treatment effectiveness

CHOICE

EPIDEMIOLOGY & EVOLUTION

PERSON STOP

Pharmacology Unit

Effectiveness is falling and the pharmacology unit has a limited budget. Every failure so far happened on documented dosing with concentrations inside the therapeutic range.

ASSUMES a matched comparison is what turns a finding into evidence

One thing before anybody's prescription changes. Which?

✓ **Sequence the failures alongside matched successful cases.**

Measure drug exposure and adherence objectively.

Increase the dose for every patient on the ward.

Test combinations and alternative targets under a clinical protocol.

ANSWER

Sequence the failures alongside matched successful cases.

WHY

Every failure here occurred on documented dosing with drug concentrations inside the therapeutic range, so the drug arrived. What is in question is whether the target site changed — and the matched successes are the control that turns a resistance-associated variant into evidence rather than a coincidence.

WHY THE OTHERS DO NOT HOLD

- × Exposure and adherence answer whether the drug ever reached the target, which is the right question when concentrations are unknown. Here they were measured and they were adequate.
- × Raising the dose for everyone treats resistance as established and exposes the three patients in four who are responding to unnecessary toxicity.
- × Combinations and alternative targets are the response once you know what changed. Chosen now, they are a guess with a protocol attached.

A resistance response should combine evolutionary evidence with pharmacology and patient safety.

MISSION 12

Design the Intervention

Select a layered intervention portfolio and identify the evidence required before deployment.

Choosing one glamorous intervention and neglecting all others could leave the city unprotected during development or against biological change.

12.1 Choose the intervention layer

PROTOCOL

EPIDEMIOLOGY & EVOLUTION

Vaccine Design Lab

The city wants an intervention and the room is arguing as though only one can be chosen. Four proposals are on the table and each acts at a different moment in the course of an infection.

ASSUMES an infection has a before, a during and an after, and different tools act at each

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

CORRECT MATCHING

Prevent initial cell entry.	→ Entry-blocking antibody or receptor-targeted strategy.
Shorten severe inflammatory disease after infection.	→ Stage-specific anti-inflammatory treatment with monitoring.
Create immune memory before exposure.	→ Vaccination.
Reduce opportunities for transmission while biomedical tools are limited.	→ Ventilation, isolation, masking, or contact reduction as appropriate.

WHY

Each class acts at a different point, which is why a layered strategy is stronger than a bigger version of any one of them. Blocking entry acts before infection is established. An anti-inflammatory acts after it is, on the damage rather than the agent. Immune memory created before exposure changes who can be infected at all. And reducing contact works on the population rather than the patient, which is why it keeps working while the biology is still unknown. Different mechanisms means different failure modes.

WHY THE OTHERS DO NOT HOLD

- × Preventing entry is a job for something that blocks the receptor interaction, and it has to be there before the agent is.
- × Shortening severe inflammatory disease is treatment after infection, and it is staged — given at the wrong moment it suppresses a response that is still needed.
- × Immune memory before exposure is vaccination, and its whole value is that it arrives first.
- × Ventilation, isolation and contact reduction act on transmission rather than on any patient, which is why they work whatever the agent turns out to be.

A layered strategy is stronger when each intervention has a defined mechanism and failure mode.

12.2 From candidate to justified trial

SEQUENCE

EPIDEMIOLOGY & EVOLUTION

PERSON STOP

Therapeutics Group

Two compounds show activity in cell culture and the city wants one of them in patients within the month. Neither has been through an animal model, and the protocol on the table names no endpoint and no stopping rule.

ASSUMES activity in a dish is not evidence of benefit in a person

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

CORRECT ORDER

- 1 Define the biological target and intended mechanism.
- 2 Demonstrate relevant activity and identify major safety concerns in preclinical evidence.
- 3 Specify trial population, endpoints, controls, and stopping rules.
- 4 Begin a monitored clinical trial with independent oversight.

WHY

Urgency changes the speed of this path and not its structure. A target and a mechanism come first because they say what the compound is supposed to do; preclinical evidence says whether it does anything and what it does to the animal; the trial design says what would count as working and when to stop. Skip the endpoint and the stopping rule and you have a trial that cannot conclude anything and cannot be halted when it should be.

WHY THE OTHERS DO NOT HOLD

- ✗ Target and mechanism first: without them there is nothing to test a compound against except hope.
- ✗ Preclinical activity and safety second, because a compound that harms an animal has told you something you needed before a person took it.
- ✗ Endpoints, controls and stopping rules third — they are what make the result interpretable, and they cannot be chosen after the data arrives.
- ✗ The monitored trial last, with oversight. It is the step everything before it exists to make defensible.

Urgency changes speed and coordination, not the need for interpretable evidence and participant protection.

12.3 Build a layered intervention portfolio

CHOICE

EPIDEMIOLOGY & EVOLUTION

PERSON STOP

Public Health Strategy Room

One portfolio to fund, several plausible layers competing for it, and a city that would like to be told the crisis is under control.

ASSUMES two tools that fail for the same reason fail at the same time

The city can start one layer this month. Which?

✓ **Ventilation and rapid-isolation improvements.**

A vaccine candidate against a conserved surface feature.

A therapeutic aimed at a validated host or agent pathway.

A publicity campaign stating that the crisis is under control.

ANSWER

Ventilation and rapid-isolation improvements.

WHY

Layers are chosen for mechanism and for timing. Ventilation and isolation reduce transmission opportunities whatever the biology turns out to be, and they work this month — which is the month the other two spend in development.

WHY THE OTHERS DO NOT HOLD

- ✗ The vaccine is the strongest long-run layer and protects nobody during the months it takes to make. Start it too; it is not what this month buys.
- ✗ A therapeutic depends on a pathway that is still being validated, and it acts after infection rather than reducing how much of it there is.
- ✗ A campaign announcing control is not a layer. It has no mechanism, and it fails in exactly the way that costs the most.

Layers are chosen for different mechanisms and different timescales, not for how promising each one sounds.

MISSION 13

The Trial Signal

Make a conditional trial decision that weighs benefit, uncertainty, and safety rather than relying on one p-value or headline percentage.

Expanding an unsafe intervention harms participants; stopping too early can discard the only effective tool.

13.1 Protect the comparison

PROTOCOL

CLINICAL PHYSIOLOGY

Clinical Trial Center

Early trial results look encouraging. Four things about how the trial was run are on the table at the clinical trial centre, and each of them could produce an encouraging number on its own.

ASSUMES a treated group is only informative next to a group that was not treated

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

CORRECT MATCHING

Sicker patients preferentially receive the candidate. → Randomization or careful allocation procedures.

Outcome assessors know who received treatment. → Blinding where feasible.

Many outcomes are tested and only the best is reported. → Pre-specified primary outcomes and multiplicity control.

Participants leave the study at different rates by group. → Track attrition and analyze its potential bias.

WHY

A trial is a machine for producing one comparison — what happened, against what would have happened otherwise — and each of these four threatens that comparison in its own way. Sicker patients getting the candidate makes the groups different before treatment starts. An assessor who knows the arm reads outcomes differently. Testing many outcomes and reporting the best finds a difference by looking often enough. And patients leaving unevenly changes who is left to measure. Each has a specific design control, and none substitutes for another.

WHY THE OTHERS DO NOT HOLD

- ✗ Sicker patients receiving the candidate is confounding by indication, and allocation is what stops it — randomisation makes the groups comparable before anything is given.
- ✗ An assessor who knows the arm is measurement bias, and blinding is the answer to it. Randomising will not help once the outcome is being judged.
- ✗ Many outcomes tested and the best reported is multiplicity: with enough outcomes something is always significant, and pre-specification is what closes it.
- ✗ Uneven attrition changes who remains to be measured, so it is tracked and analysed rather than designed away.

A credible effect depends on how the comparison was produced, not only the final number.

13.2 Read effect size and uncertainty

BALLPARK

EPIDEMIOLOGY & EVOLUTION

Biostatistics Office

The trial has reported, and the health minister wants a number before deciding how much to buy. The biostatistics office has the event counts for both arms and nothing has been converted yet.

ASSUMES a risk is events divided by the people who could have had them

Estimate the number of patients who must be treated for one to benefit.

GOVERNING RELATIONSHIP Risk = events ÷ people at risk. The number needed to treat is $1 \div (\text{control risk} - \text{treated risk})$.

20 hospitalised, control arm

200 participants, control arm

10 hospitalised, treated arm

200 participants, treated arm

30 hospitalised in the trial

400 participants in the trial

$1 \div (20 \text{ hospitalised, control arm} \div 200 \text{ participants, control arm} - 10 \text{ hospitalised, treated arm} \div 200 \text{ participants, treated arm})$

ANSWER

Risks of 0.10 and 0.05, an absolute reduction of 0.05, so about 20.

WHY

The same result can be told three ways. Risk halved sounds enormous. Five fewer hospital admissions in every hundred sounds modest. Twenty patients treated for one to benefit sounds like work. All three are true, and only the last two tell a minister how much of the drug to buy. Reporting the relative figure alone is the commonest way a real but small effect is oversold.

Relative and absolute effects answer different decision questions and should be reported together.

13.3 Continue, expand, or pause?

CHOICE

EPIDEMIOLOGY & EVOLUTION

PERSON STOP

Independent Safety Board

The result is promising and imprecise, and the adverse events are uneven across the arms. The safety board has to say what happens next, and enrolment is paused until it does.

ASSUMES a wider interval means less is known, not that the effect is smaller

Promising, imprecise, and the adverse events are uneven. What does the board decide?

✓ **Expand enrolment under the same pre-specified analysis.**

Declare success and stop collecting data.

Add mechanistic measurements to explain why it works.

Pause the trial until the adverse events are fully investigated.

ANSWER

Expand enrolment under the same pre-specified analysis.

WHY

The estimate is real and too wide to act on, and width is fixed by numbers rather than by argument. Keeping the pre-specified analysis is what makes the extra participants add precision instead of adding a second chance to find a positive result.

WHY THE OTHERS DO NOT HOLD

- ✗ Stopping now converts an imprecise estimate into a permanent one, and the imprecision is the whole reason the board was convened.
- ✗ Mechanism is worth knowing and does not narrow the interval the decision turns on.
- ✗ Pausing is right if the safety signal is real, and the safety board's monitoring is what establishes that — it runs alongside enrolment rather than instead of it.

A mature trial decision can be encouraging and cautious at the same time.

MISSION 14

Containment or Adaptation?

Use scenarios and trigger-based policies instead of pretending one forecast is certain.

An inflexible choice can either impose unnecessary costs or allow the health system to collapse.

14.1 Build a decision model

SEQUENCE

EPIDEMIOLOGY & EVOLUTION

Population Modeling Group

Three districts are on visibly different trajectories and the council wants one citywide policy by Friday. The modelling team has nine days of case data and a dozen parameters it can only bound.

ASSUMES a model's output is only as good as the assumptions it was given

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

CORRECT ORDER

- 1 State the decision, time horizon, and outcomes that matter.
- 2 Choose a transparent model structure and estimate inputs from data.
- 3 Run multiple plausible scenarios and sensitivity tests.
- 4 Link observable triggers to pre-agreed policy changes.

WHY

A model is a structured argument, not an oracle, and its assumptions are as much of the output as its numbers. Stating the decision first is what stops it modelling something nobody asked about — a model built without a decision answers whatever question was easiest. The structure and the inputs come next, then the range of plausible worlds rather than one, and finally the triggers, because a scenario that is not tied to an observable is a scenario nobody will ever act on.

WHY THE OTHERS DO NOT HOLD

- ✗ The decision, the horizon and the outcomes come first, or the model answers the question that was easiest to model.
- ✗ Structure and inputs second, and transparent, because a structure nobody can inspect cannot be argued with.
- ✗ Scenarios and sensitivity third: the point is the spread across plausible worlds, not the middle of it.
- ✗ Triggers last, tying a scenario to something observable. Without them the model produces a graph and no decision.

Models are structured arguments whose assumptions should remain visible.

14.2 Hospital capacity threshold

BALLPARK

CLINICAL PHYSIOLOGY

Hospital Capacity Center

The region has 120 staffed critical-care beds. Ninety are already full. Severe cases are arriving at about eight a day, and the average stay is somewhere near six days — a figure nobody has actually checked.

ASSUMES a bound is what you can say before the missing number arrives

Estimate the upper bound on time to capacity, assuming nobody is discharged.

GOVERNING RELATIONSHIP Time to fill = beds free ÷ arrivals per day.

30 open beds

8 admissions/day

120 staffed beds

90 beds occupied

6 days (average stay)

$30 \text{ open beds} \div 8 \text{ admissions/day}$

ANSWER

$30 \div 8 \approx 3.75$ days, if nobody leaves.

WHY

A bound is worth computing early because of what it shows is missing. Here it says the decision turns on a discharge rate nobody is collecting.

Simple bounds can reveal which missing data are decision-critical.

14.3 Choose an adaptive policy

CHOICE

EPIDEMIOLOGY & EVOLUTION

PERSON STOP

City Command

The outbreak is geographically uneven and the council wants a policy it can defend for a month. Three districts, three different trajectories, and one decision.

ASSUMES a policy tied to an observable can change without anybody having been wrong

A policy the council can defend for a month. Which?

✓ **Citywide measures tied to hospital and growth thresholds.**

Targeted containment wherever transmission chains are traceable.

Capacity expansion and protection of vulnerable populations.

Commit to one policy now and hold it whatever the data show.

ANSWER

Citywide measures tied to hospital and growth thresholds.

WHY

Uncertainty is not resolved by choosing more confidently. A policy tied to observable triggers agreed in advance changes when the outbreak changes, and it can be relaxed without anybody having to admit they were wrong — which is what keeps it defensible for a month.

WHY THE OTHERS DO NOT HOLD

- ✗ Targeted containment is cheap and effective where chains are traceable, and it fails silently where they are not. Nothing in it tells you it has stopped working.
- ✗ Capacity expansion protects the outcome that matters and does nothing about transmission, so it raises the ceiling without slowing the climb.
- ✗ Committing regardless of new evidence is the only option here that cannot learn, which in a month of an uneven outbreak is the one guaranteed to be wrong somewhere.

Good policy is a feedback system, not a one-time guess.

MISSION 15

The Final Briefing

Produce a claim-by-claim evidence package with explicit residual uncertainties and durable surveillance responsibilities.

A technically correct response can still fail if uncertainty, burdens, and continuing responsibilities are hidden from the people affected.

15.1 Disposition the final claims PROTOCOL EPIDEMIOLOGY & EVOLUTION

Scientific Review Hall

Four claims are going into the final report and they are not equally well supported. The review board has to decide what to say about each, and the document will outlive everybody in the room.

ASSUMES evidence comes in strengths, and a conclusion inherits the weakest one it rests on

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

CORRECT MATCHING

Multiple independent data streams support the transmission route.	→ Treat as established enough for operational planning while continuing surveillance.
A severe-risk subgroup is suggested by one small observational sample.	→ Label as provisional and collect representative confirmatory data.
An intervention lowers hospitalization with monitored safety but long-term duration is unknown.	→ Use conditionally with follow-up and duration studies.
An animal reservoir remains plausible but unconfirmed.	→ Avoid destructive ecological action; continue targeted One Health investigation.

WHY

The failure mode here is uniform treatment — publishing the weakest claim with the confidence of the strongest, or holding back the strongest because the weakest is unresolved. Multiple independent data streams agreeing is the strongest thing on the list and can carry operational planning. One small observational sample is a hypothesis worth collecting data on. A benefit shown with unknown duration can be used with follow-up attached. And a plausible unconfirmed reservoir is a reason to keep investigating, not a reason to destroy anything.

WHY THE OTHERS DO NOT HOLD

- ✗ Independent streams agreeing is the strongest evidence in the report, and treating it as provisional wastes the only thing the response can plan on.
- ✗ One small observational sample identifies a subgroup worth confirming. Published as established, it becomes policy about people nobody measured properly.
- ✗ A benefit whose duration is unknown is used conditionally, with follow-up — the uncertainty is about how long, not about whether.
- ✗ A plausible unconfirmed reservoir justifies more investigation. Acting on it destroys the evidence and may not touch the actual source.

Scientific conclusions should carry the strength and limits of their supporting evidence.

15.2 Build the public explanation

SEQUENCE

EPIDEMIOLOGY & EVOLUTION

PERSON STOP

City Command

The mayor is on air in an hour. The transmission route is settled, the animal reservoir is not, and the treatment works under conditions nobody has tested past eight weeks. The draft opens with the actions.

ASSUMES an audience judges a claim by how it was supported, once it knows

Order the briefing by how much weight each claim can carry.

CORRECT ORDER

- 1 The transmission route, which three independent data streams agree on.
- 2 The severe-risk subgroup, which one small observational sample suggests.
- 3 The animal reservoir, which remains plausible and unconfirmed.
- 4 What each of those would take to settle, and when it will be revisited.

WHY

The briefing is ordered by how much the evidence will carry, strongest first, because that is the order in which a listener can tell one claim from another. Publishing the reservoir with the confidence of the transmission route is what makes the whole document unbelievable when the reservoir turns out to be somewhere else.

A public account is ordered by the strength of its evidence, not by the order the work happened in.

15.3 Fund the post-crisis legacy

CHOICE

EPIDEMIOLOGY & EVOLUTION

PERSON STOP

Public Briefing Room

The emergency budget closes at the end of the month and anything uncommitted returns to the state. Four proposals, and the room has stopped pretending all of them survive.

ASSUMES capability that is not funded to continue stops when the emergency does

The emergency budget closes this month. What does the city commit it to?

✓ **Integrated clinical, environmental and genomic surveillance.**

Health-system resilience and workforce training.

Preserving the data, methods and after-action record.

Closing the records once the review is published.

ANSWER

Integrated clinical, environmental and genomic surveillance.

WHY

Every week of warning this outbreak got came from a signal somebody was already collecting. Surveillance is the one item here that has to run continuously to be worth anything, so it is the one a closing budget should endow rather than leave to next year's argument.

WHY THE OTHERS DO NOT HOLD

- ✗ Resilience and training decide whether detection helps, and they are the next call. They do nothing if nothing detects the next event.
- ✗ Preserving the record costs very little and should happen regardless. It is a filing decision, not what an emergency budget is for.
- ✗ Closing the records removes the only account of how any of this was measured, and the next response would start from nothing.

The final responsibility is to convert crisis knowledge into durable public capability.